



Lecture notes for Applies statistics, Probability theory

Lecture 9.

Dr. Barta Attila¹, Dr. Pecsora Sándor²

^{1,2}University of Debrecen, Faculty of informatics,
Department of Applied statistics and probability theory

CS BSc, CSE BSc



Presentation Overview

Non parametric
tests:
chi-squared test

Non parametric tests: chi-squared test
chi-squared test
Testing goodness of fit
Testing independence
Contingency table



Section Overview

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Recap: Parametric and non-parametric tests

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Parametric tests are used when the underlying distribution is known, but the parameters are not known. The underlying distribution is usually the normal distribution.



Recap: Parametric and non-parametric tests

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Parametric tests are used when the underlying distribution is known, but the parameters are not known. The underlying distribution is usually the normal distribution.

Non-parametric statistical methods are usually distribution-free, that is they do not rely on the assumption that the data are drawn from a given parametric family of probability distributions.



Non-parametric tests

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Some important goals of non-parametric statistics are the following.

- ① Testing the goodness of fit, that is testing whether a sample is drawn from a given distribution. Methods:
 - ① Chi-square test for goodness of fit
 - ② Kolmogorov-Smirnov one-sample test
- ② Testing homogeneity: testing whether two samples are drawn from the same distribution. Methods:
 - ① Chi-square test of homogeneity
 - ② Kolmogorov-Smirnov two-sample test
 - ③ Mann-Whitney U or Wilcoxon rank sum test
 - ④ Sign test
- ③ Testing independence: testing whether two samples are independent or measuring dependence of two samples. Methods:
 - ① Chi-square test of independence
 - ② Spearman's rank correlation coefficient



The chi-squared test. Multinomial distribution

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Let $A_1, \dots, A_r$ be a partition of the sample space.

Let $p_i = P(A_i) > 0$, $i = 1, \dots, r$.

Repeat the experiment N -times independently.

Denote by X_i the number of occurrences of A_i .

Then the joint distribution of $(X_1, \dots, X_r)$ is multinomial with parameters N and $p_1, \dots, p_r$, that is

$$P(X_1 = k_1, \dots, X_r = k_r) = \frac{N!}{k_1! \dots k_r!} p_1^{k_1} \dots p_r^{k_r},$$

where $k_1, \dots, k_r$ are non-negative integers with $k_1 + \dots + k_r = N$.

Obviously, X_i is binomial with parameters N and p_i , that is

$$\begin{aligned} P(X_i = k_i) &= \binom{N}{k_i} p_i^{k_i} (1 - p_i)^{N - k_i}, \quad k_i = 0, \dots, N, \\ \mathbb{E}X_i &= Np_i, \quad \text{Var}X_i = Np_i(1 - p_i). \end{aligned}$$



The chi-squared test. Testing goodness of fit

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Let $A_1, \dots, A_r$ be a partition of the sample space.

Assume $p_1 > 0, \dots, p_r > 0$ are given with $\sum_{i=1}^r p_i = 1$.

Decide if the null-hypothesis

$$H_0 : P(A_i) = p_i, \quad i = 1, \dots, r,$$

is true.

Repeat the experiment containing the events $A_1, \dots, A_r$ N -times independently.

The event A_i occurred k_i -times out of the N repetitions. Calculate

$$\chi^2 = \sum_{i=1}^r \frac{(k_i - Np_i)^2}{Np_i}$$

If H_0 is true, then the distribution of χ^2 is asymptotically χ_{r-1}^2 .



The chi-squared test. Testing goodness of fit

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Concept question 1.

Test if a coin is fair. So

$$H_0 : P(H) = \frac{1}{2}, \quad P(T) = \frac{1}{2}$$

Toss the coin 200 times. We obtained 110 heads, and 90 tails.



The chi-squared test. Testing goodness of fit

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The structure of the χ^2 statistic is

$$\sum \frac{(\text{observed value} - \text{expected value})^2}{\text{expected value}}.$$

So χ^2 is small if and only if the observed values are close to the values which are expected if H_0 is true.

So we accept H_0 if χ^2 is small.



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So we accept H_0 if χ^2 is small.

If χ^2 is larger than the critical value, then we reject H_0 .

For given level α the critical value is obtained from a table of χ^2_{r-1} distribution.

The χ^2 test is applicable for large N .

If there are s unknown parameters, then we estimate them from the sample. But then the degree of freedom will be $r - 1 - s$.



The chi-squared test. Testing goodness of fit

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Concept question

Test if a coin is fair. So

$$H_0 : P(H) = \frac{1}{2}, \quad P(T) = \frac{1}{2}$$

Toss the coin 200 times. We obtained 110 heads, and 90 tails.

Concept question 1. solution

$$\chi^2 = \frac{(110 - 200 \cdot 0.5)^2}{200 \cdot 0.5} + \frac{(90 - 200 \cdot 0.5)^2}{200 \cdot 0.5} = 2.$$

The critical value at significance level 10% from the table of $\chi^2_{2-1} = \chi^2_1$ distribution is 2.71.

As now $\chi^2 < 2.71$, so we accept H_0 .



The chi-squared test. Testing independence

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Let $\mathcal{A} = \{A_1, A_2, \dots, A_r\}$ and $\mathcal{B} = \{B_1, B_2, \dots, B_s\}$ be two partitions of the sample space.

Test the independence of $\mathcal{A}$ and $\mathcal{B}$, that is

$$H_0 : P(A_i \cap B_j) = P(A_i)P(B_j), \quad i = 1, \dots, r, \quad j = 1, \dots, s.$$

If the probabilities $p_i = P(A_i)$, $q_j = P(B_j)$ are known, then we have to check

$$H_0 : P(A_i \cap B_j) = p_i q_j, \quad i = 1, \dots, r, \quad j = 1, \dots, s.$$

As here

$$\{A_i \cap B_j : i = 1, \dots, r, j = 1, \dots, s\}$$

is again a partition of the sample space and

$$\{p_i \cdot q_j : i = 1, \dots, r, j = 1, \dots, s\}$$

is a given distribution, so we can follow the chi-squared test for goodness of fit.

The limit of the test statistic will be χ^2_{rs-1} .



Concept question 2.

2000 boys doing sports were involved in a research. 150 of them were gymnasts, 1500 played football, and 350 played basketball. The heights of the boys were measured. The results are in the following contingency table. Decide if the type of the sport is independent of the height.

sport \ height	< 170	170 – 190	190 <	Σ
gymnast	100	50	0	150
footballer	400	1000	100	1500
basketball player	50	100	200	350
Σ	550	1150	300	2000



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Contingency table (also known as a cross tabulation) displays the frequencies of the variables.

In applications p_i and q_j are not known. So we shall estimate them.

Repeat the experiment N -times. Denote by k_{ij} the frequency of $A_i \cap B_j$.

	B_1	B_2	$\dots$	B_s	Σ
A_1	k_{11}	k_{12}		k_{1s}	$k_{1.}$
A_2	k_{21}	k_{22}		k_{2s}	$k_{2.}$
$\vdots$					
A_r	k_{r1}	k_{r2}		k_{rs}	$k_{r.}$
Σ	$k_{.1}$	$k_{.2}$		$k_{.s}$	N



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On the margins of the table we have

$$k_{i.} = \sum_{j=1}^s k_{ij} \quad (\text{the frequency of } A_i)$$

$$k_{.j} = \sum_{i=1}^r k_{ij} \quad (\text{the frequency of } B_j)$$

We estimate the probabilities by relative frequencies:

$$p_i = P(A_i) \quad \text{is estimated by } \frac{k_{i.}}{N}$$

$$q_j = P(B_j) \quad \text{is estimated by } \frac{k_{.j}}{N}$$



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So the observed value of $A_i \cap B_j$ is k_{ij} , and its expected value (under H_0) is

$$N \cdot \frac{k_{i.}}{N} \cdot \frac{k_{.j}}{N} = \frac{k_{i.} k_{.j}}{N}$$

So the χ^2 statistics is

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^s \frac{\left(k_{ij} - \frac{k_{i.} k_{.j}}{N}\right)^2}{\frac{k_{i.} k_{.j}}{N}}.$$

The unknown parameters are $p_1, \dots, p_r, q_1, \dots, q_s$.

Because of $\sum p_i = 1$ and $\sum q_j = 1$ we have to estimate $r - 1$ values of p_i and $s - 1$ values of q_j .

So if H_0 is true, then χ^2 is asymptotically $\chi^2_{rs-1-(r+s-2)} = \chi^2_{(r-1)(s-1)}$.



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Concept question 2. solution

$$\begin{aligned}\chi^2 = & \frac{\left(100 - \frac{550 \cdot 150}{2000}\right)^2}{\frac{550 \cdot 150}{2000}} + \frac{\left(50 - \frac{1150 \cdot 150}{2000}\right)^2}{\frac{1150 \cdot 150}{2000}} + \frac{\left(0 - \frac{300 \cdot 150}{2000}\right)^2}{\frac{300 \cdot 150}{2000}} + \\ & + \frac{\left(400 - \frac{550 \cdot 1500}{2000}\right)^2}{\frac{550 \cdot 1500}{2000}} + \frac{\left(1000 - \frac{1150 \cdot 1500}{2000}\right)^2}{\frac{1150 \cdot 1500}{2000}} + \frac{\left(100 - \frac{300 \cdot 1500}{2000}\right)^2}{\frac{300 \cdot 1500}{2000}} + \\ & + \frac{\left(50 - \frac{550 \cdot 350}{2000}\right)^2}{\frac{550 \cdot 350}{2000}} + \frac{\left(100 - \frac{1150 \cdot 350}{2000}\right)^2}{\frac{1150 \cdot 350}{2000}} + \frac{\left(200 - \frac{300 \cdot 350}{2000}\right)^2}{\frac{300 \cdot 350}{2000}} = 700.72\end{aligned}$$

The degree of freedom of the chi-square distribution is $(r-1)(s-1) = 4$.
The value of 700.72 is so large that we reject independence at any usual significance level.